

Where Today's BI Tools Stop

A large, fast-growing market — still solving only half the problem.

\$40B → \$120B

Global BI market, 2026–2034 (~10–15% CAGR). A large, still-growing spend base.

~70% vs 30%

Analyst time spent preparing & investigating data vs. actually deciding (Forrester).

“Agentic shift”

Analysts (Technavio, 2026) note BI moving from static dashboards toward autonomous, reasoning-driven analytics — but today's copilots still return one flat narrative.

CAPABILITY MAP — WHAT EACH CATEGORY OF TOOL ACTUALLY DELIVERS

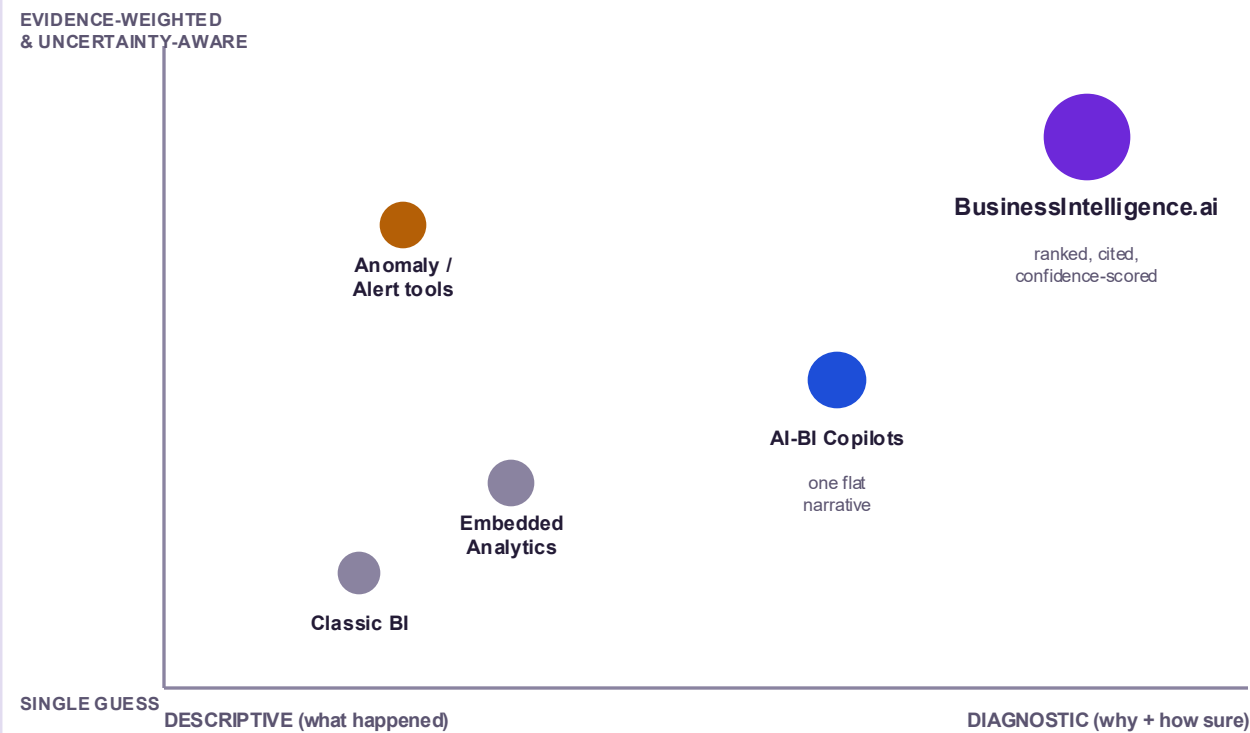
Category	Metrics & Alerts	Natural-Language Q&A	Root-Cause Ranking	Confidence & Uncertainty	What-If ₹ Impact	Closed-Loop Verification
Classic BI Power BI, Tableau, Looker	✓	✗	✗	✗	✗	✗
Embedded Analytics Sisense, Domo	✓	~	✗	✗	✗	✗
AI-BI Copilots Spotter, Tableau Pulse, MS Copilot	✓	✓	~	✗	✗	✗
Anomaly / Alert Tools Anomalo, Metabase alerts	✓	✗	~	✗	✗	✗
BusinessIntelligence.ai (this product)	✓	✓	✓	✓	✓	✓

Sources (directional estimates): Straits Research BI Market Report 2026; Forrester data-workflow research via TDWI/Tellius; Technavio BI Market Growth Analysis 2026–2030.

The Gap We Decided To Attack

Not another dashboard, not another single-answer copilot — the evidence-weighted, uncertainty-aware diagnosis layer.

POSITIONING — DESCRIPTIVE vs. DIAGNOSTIC, SINGLE-GUESS vs. EVIDENCE-WEIGHTED



WHY THIS ANGLE, WHY NOW

FEASIBLE NOW

LLMs can reliably turn scattered evidence into structured, cited reasoning — this was not credible even 18 months ago.

UNDERSERVED BUYER

Mid-market & SMB teams feel dashboard fatigue but can't hire a full analyst bench to chase every anomaly.

STRUCTURALLY DIFFERENT

A confidence-scored evidence graph + a self-challenging reverse pass can't be faked by a single prompt-wrapper.

COMPOUNDS OVER TIME

Every verified or rejected recommendation becomes training signal — confidence calibration improves per customer.

DELIBERATELY OUT OF SCOPE (v1): Not a new charting engine. Not a Power BI / Sheets replacement — we connect on top. Not autonomous action — we recommend, a human approves.

How We Turn Data Into a Diagnosis

The build methodology, from raw KPI feed to a verified, evidence-backed answer.

0 FOUNDATION

Connect data sources (KPI tables, tickets, reviews); map schema & business glossary.

1 DETECT + RETRIEVE

Statistical anomaly detection; hybrid evidence retrieval across structured + unstructured sources.

2 REASON + ALLOCATE

Build the reasoning graph; allocate a 100% contribution pool; score confidence per cause.

3 SIMULATE + RECOMMEND

Counterfactual what-if modeling; rank actions by estimated ₹ recovery.

4 VERIFY + LEARN

Track the KPI after the action; recalibrate confidence using the outcome.

STATISTICAL DETECTION

- STL decomposition + z-score / changepoint detection separates real anomalies from seasonal drift.
- Auto-scopes the investigation window (which dates, which region/product slice).

EVIDENCE RETRIEVAL

- Hybrid search (BM25 + embeddings) across tickets, reviews, ops logs, and structured KPI deltas.
- Every retrieved snippet is tagged with an Evidence ID (E1, E2 ...) for traceability.

STRUCTURED REASONING

- LLM allocates a 100% contribution pool across causes via a constrained JSON schema, citing evidence IDs.
- A second “reverse” pass actively tries to falsify the top cause before it's shown to the user.

BUILT FOR THIS ROUND

KPI ingestion (CSV / Postgres) • statistical anomaly detector • evidence retrieval over sample tickets & reviews • reasoning graph UI (share % + confidence) • what-if calculator (elasticity model) • Don't-Know mode.

STRETCH / ROADMAP

Live connectors (Shopify, Salesforce, Zendesk) • automated post-action verification against KPIs • multi-region / multi-tenant support • Slack / email daily-digest agent.

One live investigation, from alert to nested evidence to a verified fix.

app.businessintelligence.ai/investigation/8214

Revenue (Jul vs. Jun, North India)
-8.2%

Orders (Jul vs. Jun, North India)
-6.0%

Conversion (Jul vs. Jun, North India)
-5.1%

⚠️ ANOMALY DETECTED

Revenue -8.2% is 3.1σ below seasonal baseline for North India — outside normal drift.

→ INVESTIGATE ROOT CAUSE

NESTED EVIDENCE GRAPH — CAUSE → SUPPORTING EVIDENCE

Delivery delays
34% share · 82% conf

Support tickets +142% mentioning "late delivery" (E1)

Ops incident log: courier SLA breach, Aug 5 (E4)

Inventory gaps
26% share · 74% conf

In-stock rate ↓15% for top-20 SKUs (E2)

Warehouse report: restock delay, week 2 (E5)

Price / promo shift
18% share · 61% conf

🔔 Avg. price ↑7% after promo rollback (E3)

New competition
14% share · 48% conf

🚗 Rival launched -15% offer, mid-July (E6)

SIMULATE → ACT → VERIFY

WHAT-IF: Delivery time 5d → 2d

+₹35L/mo

est. recovery

RECOMMENDED ACTION

Prioritize delivery capacity in North India this week; escalate courier SLA breach.

VERIFY (14 DAYS LATER)

Delivery 5d → 2.6d • Conversion +3.8% • Revenue recovering ✓

⚠️ Don't-Know: 60% of the drop still splits across 2 causes.

A seven-agent pipeline, orchestrated end-to-end, on a pragmatic hackathon-ready stack.

THE AGENT PIPELINE

- 1

Scope Agent

Parses the question, sets timeline + region/product filters.

Tool: query planner + schema map
- 2

Detector Agent

Runs STL / z-score / changepoint tests to confirm a real anomaly.

Tool: statsmodels, ruptures
- 3

Evidence Retriever

Hybrid search across KPI deltas + tickets/reviews/logs; tags Evidence IDs.

Tool: BM25 + embeddings (pgvector)
- 4

Reasoning / Judge Agent

Allocates the 100% contribution pool; scores confidence per cause.

Tool: LLM + constrained JSON schema
- 5

Simulator Agent

Runs counterfactual what-if scenarios; estimates ₹ impact.

Tool: elasticity / regression model
- 6

Narrator Agent

Writes the executive story + selects 3–4 supporting charts.

Tool: LLM + chart-spec generator
- 7

Verifier Agent

Re-checks the KPI after action; recalibrates confidence.

Tool: scheduled job + KPI diff

TECH STACK BY LAYER

- Data Layer

Postgres / Snowflake / BigQuery connectors • dbt models for KPI standardization
- Retrieval Layer

pgvector / Weaviate embeddings • Hybrid BM25 + dense retrieval
- Orchestration

LangGraph multi-agent orchestration • Async task queue (Celery + Redis)
- Reasoning / LLM

Claude / GPT-4o for synthesis • Function calling + structured JSON output
- Backend

FastAPI services • Job scheduler for verification runs
- Frontend

Next.js / React dashboard • Recharts + custom D3 graph view
- Infra

Docker containers • Deployed on Render / Vercel; Supabase for auth+DB

10 Things Nobody Else Does

Each one closes a gap the capability map on slide 1 exposed.

1

Evidence-Linked Reasoning Graph

Every cause traces back to a cited source — support ticket, log, or metric — not a vague AI narrative.

3

Confidence-Scored Causes

Every explanation ships with an evidence-strength score, not a false sense of certainty.

5

Fact vs. Inference Separation

Every claim is tagged as an observed fact or a model inference — never blurred together.

7

What-If Impact Simulator

Quantifies the ₹ recovery for each proposed fix before any action is taken.

9

Closed-Loop Verification

Tracks whether the recommended action actually moved the KPI — and recalibrates confidence.

2

100%-Contribution Allocation

Causes compete for a shared pool instead of reporting independent, double-counted percentages.

4

Don't-Know Mode

Explicitly flags ambiguity and names the missing evidence instead of forcing a single answer.

6

Counterfactual “Reverse” Check

The engine actively tries to falsify its own top explanation before showing it to you.

8

Context Normalization

Auto-checks market trend, competition, and seasonality before blaming internal ops.

10

Connector-First, Not a New BI Tool

Sits on top of Power BI, Sheets, or Postgres instead of asking teams to migrate off them.

We don't ask an LLM to caption a chart — we make it investigate, cite evidence, quantify impact, and check its own work.